Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Independent

1. Think of a eukaryotic animal cell. Name the part of that cell which holds the chromosomes.

2. Link to today: Think about diffusion. Which way do the particles go? Do they go from a higher concentration to a lower one, or the reverse?

3. Think about osmosis. There is a partially permeable membrane. A substance goes through it. What is the name of that substance?

4. A transport process uses energy to move particles. They go against a concentration gradient. What is the name of that process?

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Exit Ticket · Independent

Evidence

Plan: find out how pH affects amylase as it acts on starch. Iodine turns blue-black when it detects starch. It stays orange-brown when it does not detect starch. Today we plan the test for later. We have not yet collected any results from a reaction.

1. Explain why amylase acts on starch and can be used again.

Link the fit at the active site to specificity. Show how the shapes fit each other. Then say how the enzyme and substrate differ.

2. Name the IV, define the timed DV with units, and name one control with a reason.

The DV needs an endpoint we can see. Explain how a change in another rate factor would affect a comparison of pH.

Under the heading Exit ticket: answer the short questions in your book.